

Gravitational Pressure Field and Its Force-Density Field

A Mathematical Framework Based on ($P_g = \omega E_d$)

$$P_g = \omega E_d$$

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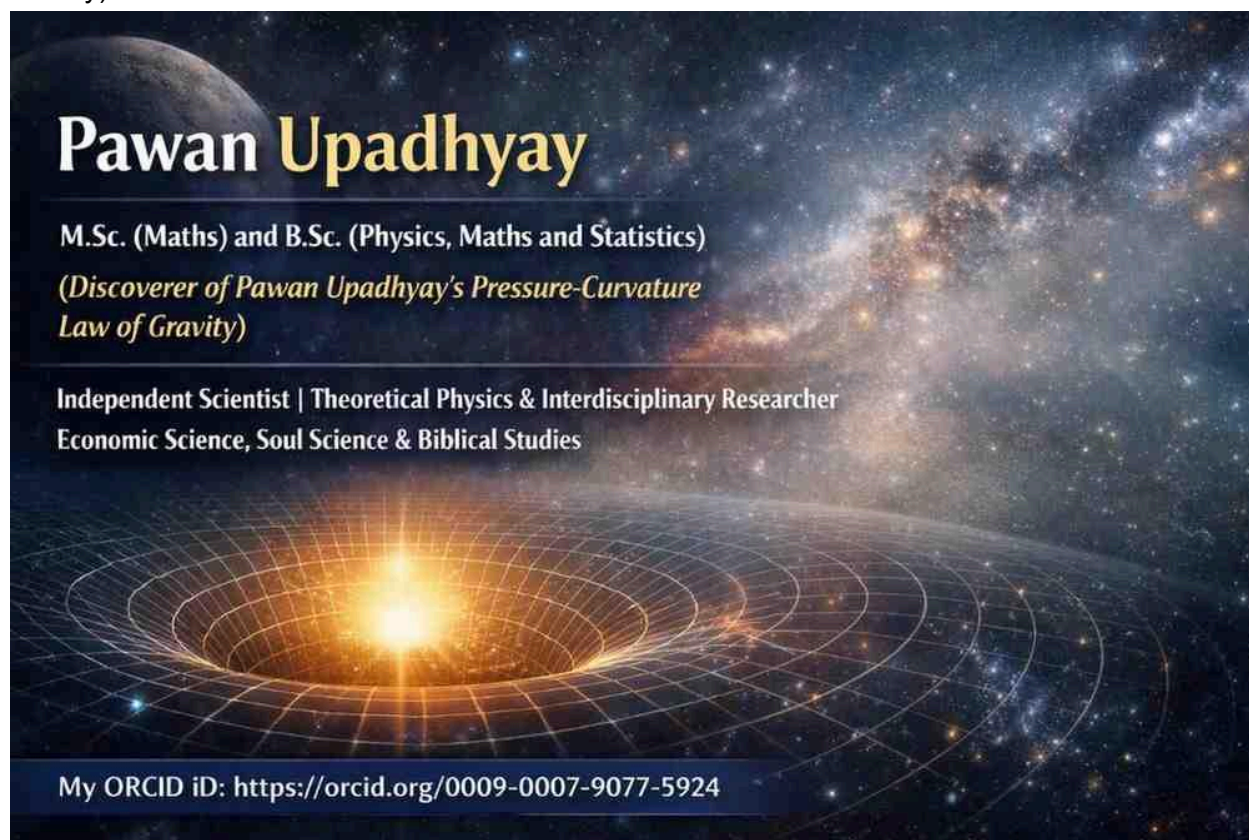
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Proposed Theory: Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)



Abstract

This paper develops a mathematical framework for describing a proposed **gravitational pressure field** (P_g) and its associated **gravitational pressure-gradient force-density field**. The central definition is

$$[\boxed{ P_g(x,y,z,ct)=\omega(x,y,z,ct), E_d(x,y,z,ct) }]$$

$$P_g(x, y, z, ct) = \omega(x, y, z, ct), E_d(x, y, z, ct)$$

where (ω) represents a scalar coupling or proportionality field and (E_d) represents an energy-density field. The spatial gradient of this pressure field generates a vector field that describes the spatial variation of gravitational pressure.

The associated force-density field is defined in this framework as

$$[\boxed{ \mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g }]$$

$$\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g$$

and is explicitly referred to throughout this paper as the **Gravitational Pressure Field Force Density**.

A central mathematical result is that the magnitude of a pressure field and the magnitude of its gradient are fundamentally different quantities. Consequently, a small value of (P_g) does not necessarily imply a small value of (∇P_g). Only when (P_g) is identically constant in a region does its spatial gradient vanish in that region.

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1. Introduction

Pressure is conventionally represented as a scalar field whose spatial variation can produce a force density. In continuum mechanics, pressure gradients play an important role in determining the direction and magnitude of pressure-induced forces.

This paper considers a proposed gravitational analogue in which a scalar field (P_g) is introduced as a **gravitational pressure field**.

The proposed field is written as

$$[\boxed{ P_g(x,y,z,ct) = \omega(x,y,z,ct) E_d(x,y,z,ct) }]$$

where the arguments

$$[(x,y,z,ct)]$$

The proposed field is written as

$$\boxed{P_g(x, y, z, ct) = \omega(x, y, z, ct) E_d(x, y, z, ct)}$$

where the arguments are

$$\boxed{(x, y, z, ct)}$$

allow the field to depend on three spatial coordinates and a temporal coordinate.

The purpose of this formulation is not to identify gravitational pressure with ordinary thermodynamic pressure, but to establish a mathematical field description in which a scalar gravitational-pressure quantity can possess spatial gradients and consequently generate an associated force-density field.

2. Definitions

2.1 Gravitational Pressure Field

The fundamental proposed definition is

$$[\boxed{P_g = \omega E_d}]$$

$$\boxed{P_g = \omega E_d}$$

or, with complete coordinate dependence,

[\boxed{ P_g(x,y,z,ct)

\omega(x,y,z,ct) E_d(x,y,z,ct) }]

$$P_g(x, y, z, ct) = \omega(x, y, z, ct), E_d(x, y, z, ct)$$

where:

- (P_g) = gravitational pressure field,
- (\omega) = Equation of state Parameter field,
- (E_d) = energy-density field.

The precise physical interpretation and dimensions of (\omega) depend on the chosen definition and unit system.

3. Gravitational Pressure Gradient Field

Because (P_g) is a scalar field, its spatial gradient is a vector field:

[\boxed{ \nabla P_g }]

with

[\nabla

\hat{\mathbf i}\frac{\partial}{\partial x} + \hat{\mathbf j}\frac{\partial}{\partial y} + \hat{\mathbf k}\frac{\partial}{\partial z}.]

Therefore,

[\boxed{ \nabla P_g

$$\frac{\partial P_g}{\partial x} \hat{\mathbf{i}} + \frac{\partial P_g}{\partial y} \hat{\mathbf{j}} + \frac{\partial P_g}{\partial z} \hat{\mathbf{k}}]$$

Because P_g is a scalar field, its spatial gradient is a vector field:

$$\boxed{\nabla P_g}$$

with

$$\boxed{\nabla = \hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}}$$

Therefore,

$$\boxed{\nabla P_g = \frac{\partial P_g}{\partial x} \hat{\mathbf{i}} + \frac{\partial P_g}{\partial y} \hat{\mathbf{j}} + \frac{\partial P_g}{\partial z} \hat{\mathbf{k}}}$$

The gradient describes how rapidly the gravitational pressure field changes from one spatial location to another.

Importantly,

$$[\boxed{P_g \neq \nabla P_g}]$$

because P_g is a scalar, whereas (∇P_g) is a vector.

$$\boxed{P_g \neq \nabla P_g}$$

because P_g is a scalar, whereas ∇P_g is a vector.

4. Derivation of the Pressure Gradient

Starting from

$$[P_g = \omega E_d,]$$

the product rule gives

$$[\boxed{\nabla P_g}$$

$$E_d \nabla \omega + \omega \nabla E_d}]$$

Starting from

$$P_g = \omega E_d,$$

the product rule gives

$$\boxed{\nabla P_g = E_d \nabla \omega + \omega \nabla E_d}$$

Thus, the gravitational pressure gradient can arise from two sources:

1. spatial variation of (ω),
2. spatial variation of (E_d).

In component form,

$$\left[\frac{\partial P_g}{\partial x} \right]$$

$$E_d \frac{\partial \omega}{\partial x} + \omega \frac{\partial E_d}{\partial x},]$$

with analogous expressions for (y) and (z).

Consequently,

$$\boxed{\nabla P_g}$$

$$E_d \nabla \omega + \omega \nabla E_d]$$

$$\frac{\partial P_g}{\partial x} = E_d \frac{\partial \omega}{\partial x} + \omega \frac{\partial E_d}{\partial x},$$

with analogous expressions for y and z .

Consequently,

$$\boxed{\nabla P_g = E_d \nabla \omega + \omega \nabla E_d}$$

is the fundamental gradient relation of the proposed model.

5. Gravitational Pressure Field Force Density

The central force-density definition of this framework is

$$[\boxed{\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g}]$$

where

$$[\boxed{\mathbf{F}_{\text{field}} = \mathbf{F}}]$$

$$\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g$$

where

$$\mathbf{F}_{\text{field}} = \mathbf{F}$$

is defined specifically as the **Gravitational Pressure Field Force Density**.

The negative sign establishes the convention that the force-density field points toward decreasing gravitational pressure.

Substitution of the pressure-gradient equation gives

$$[\boxed{ \mathbf{F} }$$

$$-E_d \nabla \omega$$

$$\omega \nabla E_d]$$

$$\mathbf{F} = -E_d \nabla \omega - \omega \nabla E_d$$

Therefore, spatial variations in either (ω) or (E_d) contribute to the gravitational pressure field force density.

6. Distinction Between Pressure and Pressure Gradient

A major mathematical principle of this framework is:

$$[\boxed{ \text{Small } P_g ; \not\rightarrow \text{small } \nabla P_g }]$$

$$\boxed{\text{Small } P_g \not\Rightarrow \text{small } \nabla P_g}$$

The value of a field at a point and the rate at which that field changes spatially are different mathematical properties.

For example, consider

$$[P_g(x) = \epsilon \sin\left(\frac{x}{\epsilon}\right),]$$

where $(0 < \epsilon \ll 1)$.

The pressure magnitude satisfies

$$[|P_g| \leq \epsilon,]$$

so the pressure can be arbitrarily small.

Nevertheless,

$$[\frac{dP_g}{dx}]$$

$$\cos\left(\frac{x}{\epsilon}\right),]$$

For Example, consider

$$P_g(x) = \epsilon \sin \left(\frac{x}{\epsilon} \right),$$

where

$$0 < \epsilon \ll 1.$$

The pressure magnitude satisfies

$$|P_g| \leq \epsilon,$$

so the pressure can be arbitrarily small.

Nevertheless,

$$\frac{dP_g}{dx} = \cos \left(\frac{x}{\epsilon} \right).$$

whose magnitude can reach unity.

Thus a small-amplitude pressure field can possess a comparatively large spatial gradient if its characteristic spatial variation is sufficiently short.

This demonstrates the general mathematical result:

[\boxed{ |P_g| \text{ small does not imply } |\nabla P_g| \text{ small}. }]

$$|P_g| \text{ small does not imply } |\nabla P_g| \text{ small.}$$

Consequently,

[\boxed{ |P_g| \text{ small does not necessarily imply } |\mathbf{F}| \text{ small}. }]

$$|P_g| \text{ small} \not\Rightarrow |\mathbf{F}| \text{ small}$$

7. The Special Case ($P_g=0$)

Care must be taken when interpreting a zero pressure value.

If

[$P_g(\mathbf{r}_0)=0$]

at only one location ($\mathbf{r}_0$), it does **not** follow that

[$\nabla P_g(\mathbf{r}_0)=0$.]

$$P_g(\mathbf{r}_0) = 0$$

at only one location $\mathbf{r}_0$, it does not follow that

$$\nabla P_g(\mathbf{r}_0) = 0.$$

For example,

$$[P_g(x) = kx]$$

has

$$[P_g(0) = 0,]$$

but

$$[\frac{dP_g}{dx} = k.]$$

Therefore,

$$[\boxed{P_g = 0 \text{ at a point}}; \not\rightarrow \nabla P_g = 0.]$$

For example,

$$P_g(x) = kx$$

has

$$P_g(0) = 0,$$

but

$$\frac{dP_g}{dx} = k.$$

Therefore,

$$\boxed{P_g = 0 \text{ at a point} \not\Rightarrow \nabla P_g = 0}$$

However, if

$$[P_g=0]$$

throughout an open region, then (P_g) is constant there and

$$[\boxed{\nabla P_g=\mathbf{0}.}]$$

Consequently,

$$[\boxed{ \mathbf{F} = \mathbf{0} }]$$

within that region under the definition $(\mathbf{F} = -\nabla P_g)$.

$$\boxed{\nabla P_g = \mathbf{0}}$$

Consequently,

$$\boxed{\mathbf{F} = \mathbf{0}}$$

within that region under the definition

$$\mathbf{F} = -\nabla P_g.$$

8. Temporal Dependence

The proposed field has the form

$$[P_g = P_g(x, y, z, ct).]$$

The spatial gradient is

$$[\nabla P_g$$

$\left(\frac{\partial P_g}{\partial x}, \frac{\partial P_g}{\partial y}, \frac{\partial P_g}{\partial z} \right).$

$$\nabla P_g = \left(\frac{\partial P_g}{\partial x}, \frac{\partial P_g}{\partial y}, \frac{\partial P_g}{\partial z} \right).$$

The temporal derivative is a separate quantity:

$\left[\frac{\partial P_g}{\partial (ct)}. \right]$

From

$[P_g = \omega E_d,]$

we obtain

$\left[\boxed{ \frac{\partial P_g}{\partial (ct)} } \right]$

$E_d \frac{\partial \omega}{\partial (ct)} + \omega \frac{\partial E_d}{\partial (ct)} \}$

$$\frac{\partial P_g}{\partial(ct)}.$$

From

$$P_g = \omega E_d,$$

we obtain

$$\boxed{\frac{\partial P_g}{\partial(ct)} = E_d \frac{\partial \omega}{\partial(ct)} + \omega \frac{\partial E_d}{\partial(ct)}}$$

Thus the complete field can possess both spatial and temporal variation.

The force-density definition proposed here depends on the spatial gradient:

$$[\boxed{\mathbf{F} = -\nabla P_g.}]$$

$$\boxed{\mathbf{F} = -\nabla P_g.}$$

9. Dimensional Considerations

A pressure has dimensions of energy per unit volume:

$$[P] = \frac{\text{energy}}{\text{volume}}.$$

$$[P] = \frac{\text{energy}}{\text{volume}}$$

A pressure gradient therefore has dimensions

$$[\nabla P]$$

$$\frac{\text{pressure}}{\text{length}}$$

$$\frac{\text{force}}{\text{volume}}.$$

$$[\nabla P] = \frac{\text{pressure}}{\text{length}} = \frac{\text{force}}{\text{volume}}$$

Hence, if

$$[\mathbf{F} = -\nabla P_g,$$

then $(\mathbf{F})$ naturally has the dimensions of **force density**:

[\boxed{ [\mathbf F]

$\frac{\text{force}}{\text{volume}}. \}$

$$\mathbf{F} = -\nabla P_g,$$

then $\mathbf{F}$ naturally has the dimensions of force density:

$$[\mathbf{F}] = \frac{\text{force}}{\text{volume}}$$

This dimensional distinction is important: the quantity $\mathbf{F}$ in this framework is a **force-density field**, not automatically a total force acting on a finite object.

For a volume (V), a corresponding total force would require integration:

[\boxed{ \mathbf{F}_{\text{total}}}

$\int_V \mathbf{F}, dV$

$-\int_V \nabla P_g, dV. \}$

$$\mathbf{F}_{\text{total}} = \int_V \mathbf{F} dV = - \int_V \nabla P_g dV$$

10. Physical Interpretation

Within the proposed framework, the hierarchy of quantities is

$$[\boxed{ P_g \quad \longrightarrow \quad \nabla P_g \quad \longrightarrow \quad \mathbf{F} }]$$

where:

$$[P_g = \text{gravitational pressure field},]$$

$$[\nabla P_g = \text{gravitational pressure-gradient field},]$$

and

$$[\boxed{ \mathbf{F}_{\text{field}} } \equiv \mathbf{F}$$

$$- \nabla P_g]$$

$$\boxed{P_g \longrightarrow \nabla P_g \longrightarrow \mathbf{F}}$$

where:

P_g = gravitational pressure field,

∇P_g = gravitational pressure-gradient field,

and

$$\boxed{\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g}$$

is the **Gravitational Pressure Field Force Density**.

The magnitude of (P_g) describes the local value of the proposed pressure field, whereas (∇P_g) describes its spatial variation.

Therefore, the gravitational force-density field in this formulation is controlled by the **spatial structure** of the pressure field rather than solely by its local magnitude.

11. Proposed Field Equations

The framework can be summarized by the following equations.

Gravitational pressure field

$$[\boxed{ P_g(x,y,z,ct)$$

$$\omega(x,y,z,ct)E_d(x,y,z,ct) \}]$$

Gravitational pressure gradient field

$$[\boxed{ \mathbf{G}_g$$

$$\nabla P_g \}]$$

or explicitly,

$$[\boxed{ \mathbf{G}_g$$

$$E_d\nabla\omega+\omega\nabla E_d \}]$$

Gravitational pressure field force density

$$[\boxed{ \mathbf{F}_{\text{field}} \equiv \mathbf{F}$$

$$-\nabla P_g \}]$$

and therefore

$$[\boxed{ \mathbf{F}$$

$$-E_d\nabla\omega-\omega\nabla E_d. \}]$$

Gravitational pressure field

$$P_g(x, y, z, ct) = \omega(x, y, z, ct) E_d(x, y, z, ct)$$

Gravitational pressure gradient field

$$\mathbf{G}_g = \nabla P_g$$

or explicitly,

$$\mathbf{G}_g = E_d \nabla \omega + \omega \nabla E_d$$

Gravitational pressure field force density

$$\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g$$

and therefore,

$$\mathbf{F} = -E_d \nabla \omega - \omega \nabla E_d$$

12. Limiting Cases

Case 1: Constant (ω)

If

$$[\nabla\omega=0,]$$

then

$$[\boxed{\mathbf{F}=-\omega\nabla E_d.}]$$

The force density is determined entirely by the spatial gradient of the energy-density field.

Case 1: Constant ω

If

$$\nabla\omega = 0,$$

then

$$\boxed{\mathbf{F} = -\omega \nabla E_d.}$$

The force density is determined entirely by the spatial gradient of the energy-density field.

Case 2: Constant (E_d)

If

$$[\nabla E_d = 0,]$$

then

$$[\boxed{\mathbf{F} = -E_d \nabla \omega.}]$$

The force density is determined by spatial variation of the equation of state parameter field.

Case 2: Constant E_d

If

$$\nabla E_d = 0,$$

then

$$\boxed{\mathbf{F} = -E_d \nabla \omega.}$$

The force density is determined by spatial variation of the equation of state parameter field.

Case 3: Both are spatially constant

If

$$[\nabla \omega = 0, \quad \nabla E_d = 0,]$$

then

$$[\boxed{\mathbf{F} = 0.}]$$

Case 3: Both are spatially constant

If

$$\nabla \omega = 0, \quad \nabla E_d = 0,$$

then

$$\boxed{\mathbf{F} = 0.}$$

Case 4: (P_g) is small but rapidly varying

A small (P_g) can still produce a substantial

$$[\nabla P_g]$$

if the characteristic spatial scale of variation is sufficiently small.

Thus,

$$[\boxed{P_g \approx 0 \not\rightarrow \mathbf{F} \approx 0.}]$$

The stronger condition

$$[\boxed{P_g \equiv 0}]$$

throughout a region does imply

$$[\boxed{\mathbf{F} = 0}]$$

in that region.

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in that region.

13. Conceptual Significance

The distinction between field magnitude and field gradient is essential for any theory based on spatially varying fields.

A scalar field can be small while possessing a large derivative. Consequently, observations or calculations involving only the local value of (P_g) are insufficient to determine the magnitude of the associated force-density field.

The proposed relation

$$[\boxed{\mathbf{F} = -\nabla(\omega E_d)}]$$

$$\mathbf{F} = -\nabla(\omega E_d)$$

makes this dependence explicit.

Expanding it,

$$[\boxed{\mathbf{F}}$$

$$-E_d \nabla \omega - \omega \nabla E_d.]]$$

$$\mathbf{F} = -E_d \nabla \omega - \omega \nabla E_d$$

This equation suggests that the force-density structure depends on both the energy-density distribution and the spatial behavior of the equation of state parameter field.

14. Limitations and Relation to Established Physics

The equations presented here constitute a **proposed mathematical framework** based on the definitions introduced in this paper. The notation

$$[P_g = \omega E_d]$$

and

$$[\mathbf{F}_{\text{field}} = \mathbf{F} = -\nabla P_g]$$

$$P_g = \omega E_d$$

and

$$\boxed{\mathbf{F}_{\text{field}} = \mathbf{F} = -\nabla P_g}$$

should therefore not be interpreted as established equations of general relativity or as a replacement for the Einstein field equations without further theoretical development and empirical validation.

15. Conclusion

This paper establishes a mathematical formulation for a proposed gravitational pressure field:

$$[\boxed{ P_g(x,y,z,ct) = \omega(x,y,z,ct) E_d(x,y,z,ct). }]$$

$$P_g(x, y, z, ct) = \omega(x, y, z, ct) E_d(x, y, z, ct)$$

Its spatial gradient is

$$\boxed{\nabla P_g}$$

$$E_d \nabla \omega + \omega \nabla E_d.$$

$$\boxed{\nabla P_g = E_d \nabla \omega + \omega \nabla E_d}$$

The associated force-density field is defined as

$$\boxed{\mathbf{F}_{\text{field}} \equiv \mathbf{F}}$$

$$-\nabla P_g,$$

$$\boxed{\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g}$$

where $\mathbf{F}$ is explicitly designated as the **Gravitational Pressure Field Force Density**.

The central mathematical conclusion is

$$\boxed{P_g \not\rightarrow \nabla P_g.}$$

Likewise,

[\boxed{ P_g \text{ small}; \not\rightarrow; \mathbf{F} \text{ small}. }]

Only when (P_g) is identically constant—particularly $(P_g \equiv 0)$ —throughout a region does the corresponding pressure-gradient field vanish:

[\boxed{ P_g \equiv 0 \quad \rightarrow \quad \nabla P_g = 0 \quad \rightarrow \quad \mathbf{F} = 0. }]

$$P_g \text{ small} \not\Rightarrow \nabla P_g \text{ small.}$$

Likewise,

$$P_g \text{ small} \not\Rightarrow \mathbf{F} \text{ small.}$$

Only when P_g is identically constant—particularly $P_g \equiv 0$ —throughout a region does the corresponding pressure-gradient field vanish:

$$P_g \equiv 0 \quad \Rightarrow \quad \nabla P_g = 0 \quad \Rightarrow \quad \mathbf{F} = 0.$$

Thus the proposed framework separates three conceptually different quantities:

[\boxed{ \text{Gravitational Pressure} ; P_g }]

[\boxed{ \text{Gravitational Pressure Gradient} ; \nabla P_g }]

Gravitational Pressure ; P_g

Gravitational Pressure Gradient ; ∇P_g

[\boxed{ \text{Gravitational Pressure Field Force Density} ; \mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g. }]

Gravitational Pressure Field

Force Density

$$\mathbf{F}_{\text{field}} \equiv \mathbf{F} \\ = -\nabla P_g$$

This distinction provides a mathematically consistent starting point for further investigation of the proposed gravitational-pressure model.

Equation 1 – Gravitational Pressure Field

$$P_g(x, y, z, ct) = \omega(x, y, z, ct) E_d(x, y, z, ct)$$

Equation 2 – Gravitational Pressure Gradient Field

$$\nabla P_g = E_d \nabla \omega + \omega \nabla E_d$$

Equation 3 – Gravitational Pressure Field Force Density

$$\mathbf{F}_{\text{field}} \equiv \mathbf{F} = -\nabla P_g$$

Equation 4 – Expanded Force-Density Relation

$$\mathbf{F} = -E_d \nabla \omega - \omega \nabla E_d$$

Equation 5 — Total Force from Force Density

$$\mathbf{F}_{\text{total}} = \int_V \mathbf{F} dV$$

**Gravitational Pressure Field and Its Force Density Field
With Equation-of-State Parameter ω**

1. Gravitational Pressure Field (Definition)

$$P_g(x, y, z, ct) = \underbrace{\omega(x, y, z, ct)}_{\text{Equation-of-State Parameter (EoS)}} \underbrace{E_d(x, y, z, ct)}_{\text{Energy-Density Field}}$$

Gravitational Pressure Field Equation-of-State Parameter (EoS) Energy-Density Field

Where:
 $P_g(x, y, z, ct)$: Gravitational pressure field (scalar)
 $\omega(x, y, z, ct)$: Equation-of-state parameter (scalar, dimensionless)
 $E_d(x, y, z, ct)$: Energy-density field (scalar)
 x, y, z : Spatial coordinates
 ct : Time coordinate (in length units)

Equation of State
 $P_g = \omega E_d$
 Here, ω characterizes the relationship between gravitational pressure and energy density.
 Examples (cosmology):
 $\omega = 0$ (matter)
 $\omega = 1/3$ (radiation)
 $\omega = -1$ (vacuum energy / dark energy)

2. Gravitational Pressure Gradient Field

$$\nabla P_g(x, y, z, ct) = E_d \nabla \omega + \omega \nabla E_d$$

$$\nabla P_g = \frac{\partial P_g}{\partial x} \hat{x} + \frac{\partial P_g}{\partial y} \hat{y} + \frac{\partial P_g}{\partial z} \hat{z}$$

Gradient Operator
 $\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$

- P_g : scalar field
- ∇P_g : vector field (pressure gradient)

3. Gravitational Pressure Field Force Density

$$\mathbf{F}(x, y, z, ct) = \mathbf{F}_{\text{field}}(x, y, z, ct) = -\nabla P_g = -[E_d \nabla \omega + \omega \nabla E_d]$$

- $\mathbf{F}(x, y, z, ct)$: Gravitational pressure field force density (vector)
- $\mathbf{F}_{\text{field}}$: Same as $\mathbf{F}$ (notation to indicate field quantity)
- Direction: Toward decreasing gravitational pressure.
- Dimensions: $[\mathbf{F}]$ = force per unit volume.

4. Component Form of the Force Density Field

$$\mathbf{F}(x, y, z, ct) = F_x(x, y, z, ct) \hat{x} + F_y(x, y, z, ct) \hat{y} + F_z(x, y, z, ct) \hat{z}$$

Where:
 F_x, F_y, F_z : Components of the gravitational pressure field force density

5. Important Mathematical Result

$P_g \text{ small} \neq \nabla P_g \text{ small} \neq \mathbf{F} \text{ small}$

- A small value of gravitational pressure does not necessarily imply a small pressure gradient or a small force density.
- The gradient depends on the spatial variation of ω and E_d , not only on the magnitude of P_g .

7. Zero Pressure

(a) At a single point
 If $P_g(x_0, y_0, z_0, ct_0) = 0$
 then ∇P_g may still be non-zero.

$P_g = 0 \text{ at a point} \neq \mathbf{F} = 0$

(b) Throughout a region
 If $P_g(x, y, z, ct) \equiv 0$ in a region,
 then $\nabla P_g = 0$ and $\mathbf{F} = 0$.

$P_g \equiv 0 \Rightarrow \nabla P_g = 0 \Rightarrow \mathbf{F} = 0$

8. Characteristic Length Scale

$$|\nabla P_g| \sim \frac{|P_g|}{L}$$

Even if $|P_g|$ is small, a small spatial scale L can produce a significant pressure gradient and force density.

6. Special Cases

(i) Constant ω ($\nabla \omega = 0$): $\mathbf{F} = -\omega \nabla E_d$

(ii) Constant E_d ($\nabla E_d = 0$): $\mathbf{F} = -E_d \nabla \omega$

(iii) Both ω and E_d constant: $\nabla P_g = 0 \Rightarrow \mathbf{F} = 0$

9. Summary of Field Quantities

Symbol	Meaning
$P_g(x, y, z, ct)$	Gravitational pressure field (scalar)
$\omega(x, y, z, ct)$	Equation-of-state parameter (scalar, dimensionless)
$E_d(x, y, z, ct)$	Energy-density field (scalar)
$\nabla P_g(x, y, z, ct)$	Gravitational pressure gradient field (vector)
$\mathbf{F}(x, y, z, ct)$	Gravitational pressure field force density (vector)
$\mathbf{F}_{\text{field}}(x, y, z, ct)$	Same as $\mathbf{F}$ (field notation)

The gravitational pressure field force density is defined as $\mathbf{F}(x, y, z, ct) = -\nabla P_g(x, y, z, ct)$, with $P_g = \omega E_d$ and ω as the equation-of-state parameter.

Gravitation
as a Field of
Pressure Gradient

Gravitational Pressure Field and Its Force Density Field

With Equation-of-State Parameter ω

A Mathematical Field Framework

ω : Equation-of-State
Parameter (EoS)

1. Gravitational Pressure Field (Definition)

$$P_g(x, y, z, ct) = \omega(x, y, z, ct) E_d(x, y, z, ct)$$

Gravitational
Pressure Field
(scalar)

Equation-of-State
Parameter (EoS)
(scalar, dimensionless)

Energy-Density
Field
(scalar)

Variables and Notation

$P_g(x, y, z, ct)$: Gravitational pressure field (scalar)
 $\omega(x, y, z, ct)$: Equation-of-state parameter (EoS) (scalar, dimensionless)
 $E_d(x, y, z, ct)$: Energy-density field (scalar)
 x, y, z : Spatial coordinates
 ct : Time coordinate (in length units)
 ∇ : Gradient operator
 $\mathbf{F}(x, y, z, ct)$: Gravitational pressure field force density (vector)

What is the Equation-of-State Parameter ω ?

$$\omega = \frac{P_g}{E_d}$$

- ω characterizes the relationship between gravitational pressure and energy density.
- It is a scalar, usually dimensionless.
- Examples (in cosmology):
 - $\omega = 0$ (matter)
 - $\omega = 1/3$ (radiation)
 - $\omega = -1$ (vacuum energy / dark energy)

2. Gravitational Pressure Gradient Field

$$\nabla P_g(x, y, z, ct) = E_d \nabla \omega + \omega \nabla E_d$$

In component form:

$$\nabla P_g = \frac{\partial P_g}{\partial x} \hat{i} + \frac{\partial P_g}{\partial y} \hat{j} + \frac{\partial P_g}{\partial z} \hat{k}$$

Gradient Operator

$$\nabla = \hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z}$$

- P_g : scalar field
- ∇P_g : vector field (pressure gradient)

3. Gravitational Pressure Field Force Density

$$\mathbf{F}(x, y, z, ct) = \mathbf{F}_{\text{field}}(x, y, z, ct) = -\nabla P_g = -[E_d \nabla \omega + \omega \nabla E_d]$$

- $\mathbf{F}(x, y, z, ct)$: Gravitational pressure field force density (vector)
- $\mathbf{F}_{\text{field}}$: Same as $\mathbf{F}$ (notation to indicate field quantity)
- Direction: Toward decreasing gravitational pressure.
- Dimensions: $[F] = \text{force per unit volume (N m}^{-3}\text{)}$.

4. Component Form of the Force Density Field

$$\mathbf{F}(x, y, z, ct) = F_x(x, y, z, ct) \hat{i} + F_y(x, y, z, ct) \hat{j} + F_z(x, y, z, ct) \hat{k}$$

F_x, F_y, F_z : Components of the gravitational pressure field force density

5. Important Mathematical Result

$$P_g \text{ small} \neq \nabla P_g \text{ small} \neq \mathbf{F} \text{ small}$$

- A small value of gravitational pressure does not necessarily imply a small pressure gradient or a small force density.
- The gradient depends on the spatial variation of ω and E_d , not only on the magnitude of P_g .

7. Zero Pressure

(a) At a single point

If $P_g(x_0, y_0, z_0, ct_0) = 0$
then ∇P_g may still be non-zero.

$P_g = 0 \text{ at a point} \neq \mathbf{F} = 0$

(b) Throughout a region

If $P_g(x, y, z, ct) \equiv 0$ in a region,
then $\nabla P_g = 0$ and $\mathbf{F} = 0$.

$P_g \equiv 0 \Rightarrow \nabla P_g = 0 \Rightarrow \mathbf{F} = 0$

8. Characteristic Length Scale

$$|\nabla P_g| \sim \frac{|P_g|}{L}$$

Even if $|P_g|$ is small, a small spatial scale L can produce a significant pressure gradient and force density.

6. Special Cases

(i) Constant ω ($\nabla \omega = 0$): $\mathbf{F} = -\omega \nabla E_d$

(ii) Constant E_d ($\nabla E_d = 0$): $\mathbf{F} = -E_d \nabla \omega$

(iii) Both ω and E_d constant: $\nabla P_g = 0 \Rightarrow \mathbf{F} = 0$

9. Summary of Field Quantities

Symbol	Meaning	Type
$P_g(x, y, z, ct)$	Gravitational pressure field	Scalar
$\omega(x, y, z, ct)$	Equation-of-state parameter (EoS)	Scalar (dimensionless)
$E_d(x, y, z, ct)$	Energy-density field	Scalar
$\nabla P_g(x, y, z, ct)$	Gravitational pressure gradient field	Vector
$\mathbf{F}(x, y, z, ct)$	Gravitational pressure field force density	Vector
$\mathbf{F}_{\text{field}}(x, y, z, ct)$	Same as $\mathbf{F}$ (field notation)	Vector

From Fields to Gravitation
A New Perspective

$$\mathbf{F}(x, y, z, ct) = -\nabla P_g(x, y, z, ct), \text{ with } P_g = \omega E_d \text{ and } \omega \text{ as the equation-of-state parameter.}$$

A Proposed Mathematical Framework